

Graph (d): Blockchain Heterogeneity Across Nine Layer-1 Chains

Throughput, block-time, and finality on a log scale. Bar colour = consensus family. Chainspect snapshot 2026-06-04.

Sources (click to open)

- **Chainspect Dashboard (real-time blockchain metrics)**
<https://chainspect.app/dashboard>
 - **Spark Money - Blockchain Speed Comparison**
<https://www.spark.money/learn/blockchain-speed-comparison>
 - **The Block - On-Chain Metrics**
<https://www.theblock.co/data/on-chain-metrics>
 - **DefiLlama - Chain Statistics**
<https://defillama.com/chains>
 - **Bitcoin whitepaper (Nakamoto, 2008)**
<https://bitcoin.org/bitcoin.pdf>
 - **Ethereum Proof-of-Stake (Gasper) overview**
<https://ethereum.org/en/developers/docs/consensus-mechanisms/pos/>
 - **Solana - Proof of History**
<https://solana.com/news/proof-of-history>
 - **Cardano - Ouroboros protocol**
<https://iohk.io/en/research/library/papers/ouroboros-a-provably-secure-proof-of-stake-blo...>
 - **Polkadot - NPoS and GRANDPA**
<https://wiki.polkadot.network/learn/learn-consensus/>
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Methodology / notes

All values are taken from the Chainspect dashboard's 2026-06-04 snapshot, cross-validated against Spark Money's blockchain-speed comparison and DefiLlama chain statistics. Throughput is real (not theoretical) TPS; block time is the median inter-block interval; finality is time to economic / probabilistic finality at the chain-specific confidence threshold. Consensus-family labels are taken from each chain's official documentation or whitepaper.

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Data table (values plotted in the graph)

Chain	Throughput (TPS)	Block time (s)	Finality (s)	Consensus family
Bitcoin	4.87	672.0	3,600.0	PoW
Ethereum	22.80	12.04	768.0	PoS (Gasper)
Solana	1,700.00	0.398	12.8	PoH + PoS
BNB Chain	213.00	0.451	0.65	PoSA / DPoS
Avalanche	34.54	1.03	2.0	Avalanche BFT
Polygon	77.10	1.50	5.0	PoS (Heimdall)
Cardano	0.46	20.30	120.0	Ouroboros PoS
Polkadot	0.02	5.99	30.0	NPoS + GRANDPA
Algorand	9.09	2.73	0.5	PPoS (BFT)